

Yuxin Tang

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SKILLS

PyTorch (Megatron-LM), JAX (MaxText), TransformerEngine, HuggingFace, Triton, CuTe-DSL

EDUCATION

Ph.D. Computer Science, Rice University, 2018-2024
Advisor: Chris Jermaine
Exchange Statistics, University of California, Los Angeles, 2016-2018
B.S. Computer Science, Shanghai Jiao Tong University, 2014-2018

WORK EXPERIENCE

Amazon AGI Foundations, *Member of Technical Staff*, New York, NY 2025.01–Present

- Researched novel compute-optimal **fine-grained** MoE model scaling strategy for multimodal pretraining with hardware-aware joint optimization, analyzing the trade-offs between **TPP**, **MoE expert sparsity**, **loss-free load balancing**, **token-choice routing** and hardware features to guide architecture sparsity design. Help with loss spike diagnosis.

- Developed hardware-aware **performance model** with roofline analysis to predict training throughput across **10,000** B200 GPUs, incorporating comp-comm overlapping, 5D parallelism and **megakernels**. Boosting MFU by algorithm system co-design. (+5% MFU)

Bosch Center for Artificial Intelligence (BCAI), *Research Intern*, Sunnyvale, CA 2024.05–2024.08

- Work on an algorithmic framework for automatic prompt optimization and prompt tuning.
- Deploy prompt optimization framework to help with Bosch’s internal document queries.

Visa Research, *Research Intern*, Palo Alto, CA 2023.05–2023.08

- Design algorithms for subgraph pattern discovery within trillion-sized transaction graphs.
- Implement biclique computation framework designed to efficiently handle bipartite graphs.

PUBLICATION

- 2026 **Automated Tensor-Relational Decomposition for Large-Scale Sparse Tensor Computation.**
Yuxin Tang, Zhiyuan Xin, Zhimin Ding, Xinyu Yao, Daniel Bourgeois, Tirthak Patel, Chris Jermaine.
VLDB’26
- 2025 **EinDecomp: Decomposition of Declaratively-Specified Machine Learning and Numerical Computations for Parallel Execution.**
Daniel Bourgeois, Zhimin Ding, Dimitrije Jankov, Jiehui Li, Mahmoud Sleem, Yuxin Tang, Jiawen Yao, Xinyu Yao, Chris Jermaine.
VLDB’25
- 2024 **Relational Computation for Very Large-Scale Machine Learning.**
Yuxin Tang

- Ph.D. Thesis*
- 2024 **Monarch: Distributed Butterfly Counting for Large-scale Bipartite Graph.**
Yuxin Tang, Mangesh Bendre, Mahashweta Das.
IEEE Big Data'24
- 2024 **Soft Prompt Recovers Compressed LLMs, Transferably.**
Zhaozhuo Xu*, Zirui Liu*, Beidi Chen, Shaochen Zhong, Yuxin Tang, Jue Wang, Kaixiong Zhou, Xia Hu, Anshumali Shrivastava.
ICML'24
- 2023 **Federated Learning Over Images: Vertical Decompositions and Pre-Trained Backbones Are Difficult to Beat.**
Yuxin Tang*, Ed Hu*, Anastasios Kyrillidis, Chris Jermaine.
ICCV'23
- 2023 **Auto-Differentiation of Relational Computations for Very Large Scale Machine Learning.**
Yuxin Tang, Zhimin Ding, Dimitrije Jankov, Binhang Yuan, Daniel Bourgeois, Chris Jermaine.
ICML'23
- 2022 **Distributed learning of fully connected neural networks using independent subnet training.**
Binhang Yuan, Cameron R. Wolfe, Chen Dun, Yuxin Tang, Anastasios Kyrillidis, Chris Jermaine.
VLDB'22
- 2021 **Tensor Relational Algebra for Machine Learning System Design.**
Binhang Yuan, Dimitrije Jankov, Jia Zou, Yuxin Tang, Daniel Bourgeois, and Chris Jermaine.
VLDB'21
- 2020 **Programmable In-Network Security for Context-aware BYOD Policies.**
Qiao Kang, Lei Xue, Adam Morrison, Yuxin Tang, Ang Chen, Xiapu Luo.
USENIX Security'20
- 2018 **A Programmable, Hardware-Assisted Network Protocol Fuzzer.**
Yuxin Tang, Ang Chen.
OSDI'18 (Poster)
- 2017 **Exploring Simulation of Software-Defined Underwater Wireless Networks.**
Li Wei, Yuxin Tang, Yuching Cao, Zhaohui Wang, Mario Gerla.
MobiCom'17 Workshop on Underwater Networks

SERVICE

Conference Reviewer:

ICLR 2021–2025, ICML 2020–2025, NeurIPS 2021–2025

Session Chair:

VLDB 2023, MLSys 2025

Program Committee:

CGO 2023, UDM-AAAI 2023, PLDI 2023

Data to Knowledge (D2K) Fellow:

Fall 2022, Fall 2023, Spring 2024